

Business Insights Report (EDA)

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Internship Project

Import required libraries for data analysis and visualization

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load the marketing campaign dataset

```
df = pd.read_excel('/content/marketing_campaign.xlsx')
```

Display the first 5 rows of the dataset

```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	...	NumWebVisits
0	5524	1957	Graduation	Single	58138.0	0	0	2012-09-04	58	635	...	
1	2174	1954	Graduation	Single	46344.0	1	1	2014-03-08	38	11	...	
2	4141	1965	Graduation	Together	71613.0	0	0	2013-08-21	26	426	...	
3	6182	1984	Graduation	Together	26646.0	1	0	2014-02-10	26	11	...	
4	5324	1981	PhD	Married	58293.0	1	0	2014-01-19	94	173	...	

5 rows × 29 columns

Display dataset information including columns and data types

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 29 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   ID                                     2240 non-null   int64
1   Year_Birth                           2240 non-null   int64
2   Education                             2240 non-null   object
3   Marital_Status                       2240 non-null   object
4   Income                               2216 non-null   float64
5   Kidhome                              2240 non-null   int64
6   Teenhome                             2240 non-null   int64
7   Dt_Customer                          2240 non-null   object
8   Recency                              2240 non-null   int64
9   MntWines                             2240 non-null   int64
10  MntFruits                             2240 non-null   int64
11  MntMeatProducts                       2240 non-null   int64
12  MntFishProducts                       2240 non-null   int64
13  MntSweetProducts                     2240 non-null   int64
14  MntGoldProds                         2240 non-null   int64
15  NumDealsPurchases                    2240 non-null   int64
16  NumWebPurchases                      2240 non-null   int64
17  NumCatalogPurchases                 2240 non-null   int64
18  NumStorePurchases                   2240 non-null   int64
19  NumWebVisitsMonth                   2240 non-null   int64
20  AcceptedCmp3                        2240 non-null   int64
21  AcceptedCmp4                        2240 non-null   int64
22  AcceptedCmp5                        2240 non-null   int64
23  AcceptedCmp1                        2240 non-null   int64
24  AcceptedCmp2                        2240 non-null   int64
25  Complain                             2240 non-null   int64
26  Z_CostContact                       2240 non-null   int64
```

```
27 Z_Revenue          2240 non-null  int64
28 Response           2240 non-null  int64
dtypes: float64(1), int64(25), object(3)
memory usage: 507.6+ KB
```

✓ Check for missing values in each column

```
df.isnull().sum()
```

	0
ID	0
Year_Birth	0
Education	0
Marital_Status	0
Income	24
Kidhome	0
Teenhome	0
Dt_Customer	0
Recency	0
MntWines	0
MntFruits	0
MntMeatProducts	0
MntFishProducts	0
MntSweetProducts	0
MntGoldProds	0
NumDealsPurchases	0
NumWebPurchases	0
NumCatalogPurchases	0
NumStorePurchases	0
NumWebVisitsMonth	0
AcceptedCmp3	0
AcceptedCmp4	0
AcceptedCmp5	0
AcceptedCmp1	0
AcceptedCmp2	0
Complain	0
Z_CostContact	0
Z_Revenue	0
Response	0

dtype: int64

✓ Remove rows containing missing values

```
df = df.dropna()
```

✓ Remove duplicate rows from the dataset

```
df = df.drop_duplicates()
```

✓ Display all column names present in the dataset

df.columns

```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income', 'Kidhome',
      'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',
      'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',
      'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',
      'NumCatalogPurchases', 'NumStorePurchases', 'NumWebVisitsMonth',
      'AcceptedCmp3', 'AcceptedCmp4', 'AcceptedCmp5', 'AcceptedCmp1',
      'AcceptedCmp2', 'Complain', 'Z_CostContact', 'Z_Revenue', 'Response'],
      dtype='object')
```

Convert customer enrollment date into datetime format

```
df['Dt_Customer'] = pd.to_datetime(df['Dt_Customer'])
```

Create a new column representing total customer spending

```
df['Total_Spending'] = (
    df['MntWines'] +
    df['MntFruits'] +
    df['MntMeatProducts'] +
    df['MntFishProducts'] +
    df['MntSweetProducts'] +
    df['MntGoldProds']
)
```

Display statistical summary of numerical columns

df.describe()

	ID	Year_Birth	Income	Kidhome	Teenhome	Dt_Customer	Recency	MntWines	MntFru
count	2216.000000	2216.000000	2216.000000	2216.000000	2216.000000	2216	2216.000000	2216.000000	2216.000
mean	5588.353339	1968.820397	52247.251354	0.441787	0.505415	2013-07-10 11:29:27.509025280	49.012635	305.091606	26.356
min	0.000000	1893.000000	1730.000000	0.000000	0.000000	2012-07-30 00:00:00	0.000000	0.000000	0.000
25%	2814.750000	1959.000000	35303.000000	0.000000	0.000000	2013-01-16 00:00:00	24.000000	24.000000	2.000
50%	5458.500000	1970.000000	51381.500000	0.000000	0.000000	2013-07-08 12:00:00	49.000000	174.500000	8.000
75%	8421.750000	1977.000000	68522.000000	1.000000	1.000000	2013-12-31 00:00:00	74.000000	505.000000	33.000
max	11191.000000	1996.000000	666666.000000	2.000000	2.000000	2014-06-29 00:00:00	99.000000	1493.000000	199.000
std	3249.376275	11.985554	25173.076661	0.536896	0.544181	NaN	28.948352	337.327920	39.793

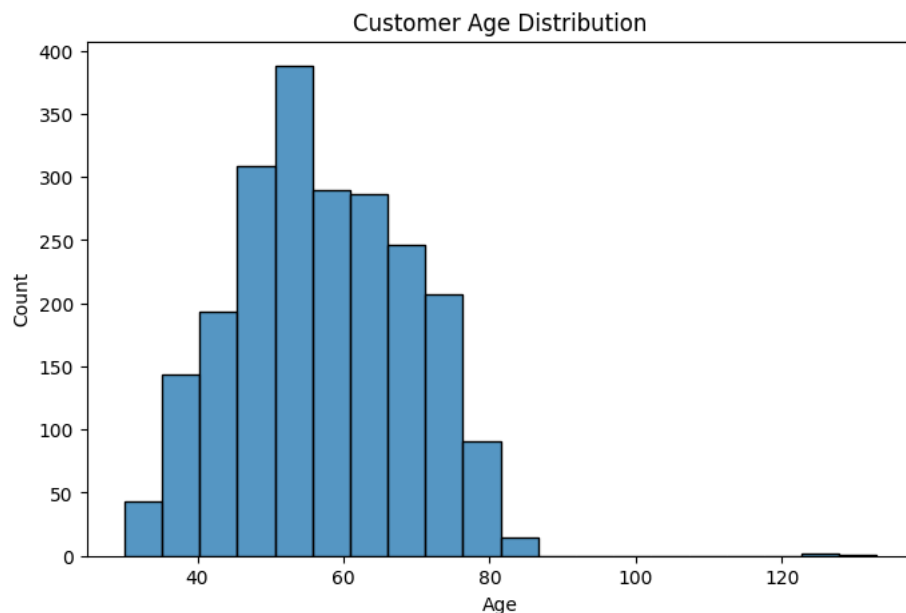
8 rows × 10 columns

Visualize customer age distribution

```
plt.figure(figsize=(8,5))
sns.histplot(2026 - df['Year_Birth'], bins=20)

plt.title('Customer Age Distribution')
plt.xlabel('Age')
plt.ylabel('Count')

plt.show()
```



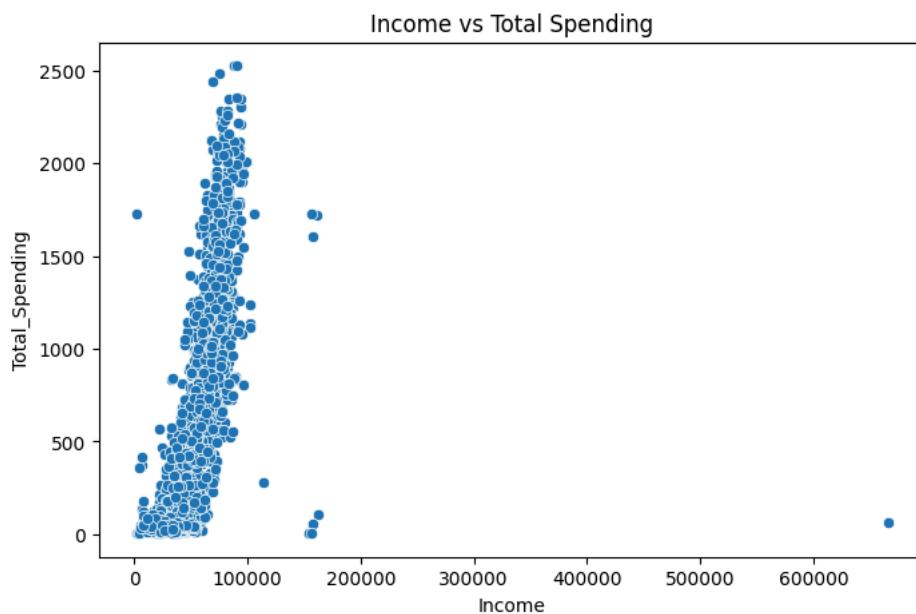
✓ Analyze relationship between customer income and spending

```
plt.figure(figsize=(8,5))

sns.scatterplot(
    x='Income',
    y='Total_Spending',
    data=df
)

plt.title('Income vs Total Spending')

plt.show()
```

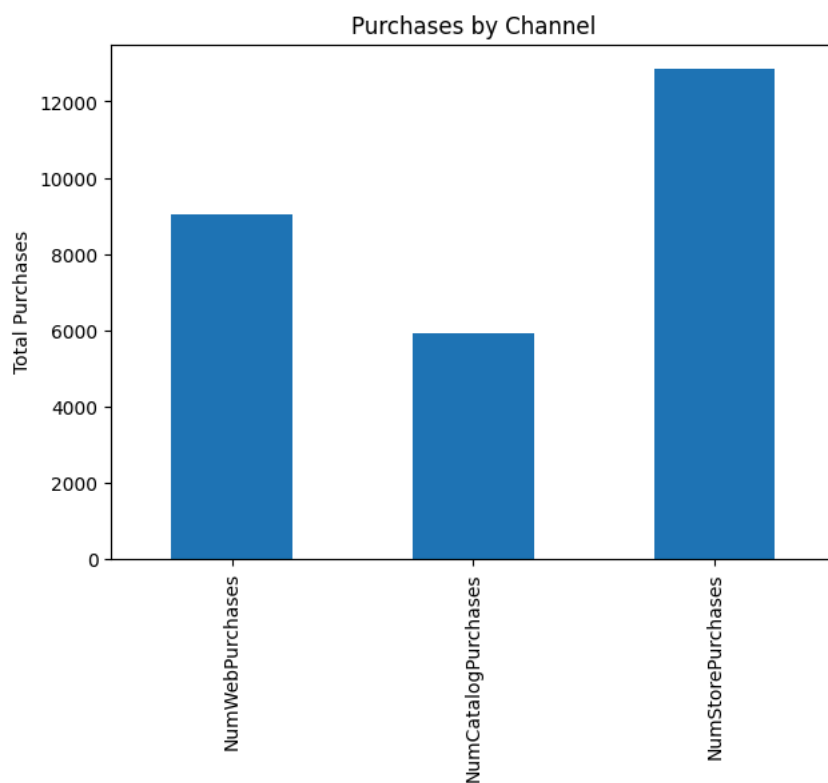


✓ Compare purchases across different marketing channels

```
channels = [
    'NumWebPurchases',
    'NumCatalogPurchases',
    'NumStorePurchases'
]

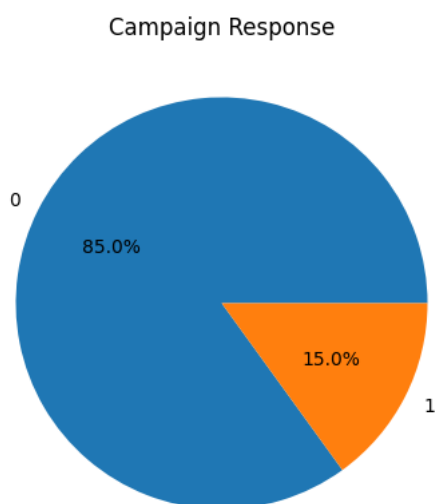
df[channels].sum().plot(
    kind='bar',
    figsize=(7,5)
```

```
)  
  
plt.title('Purchases by Channel')  
plt.ylabel('Total Purchases')  
  
plt.show()
```



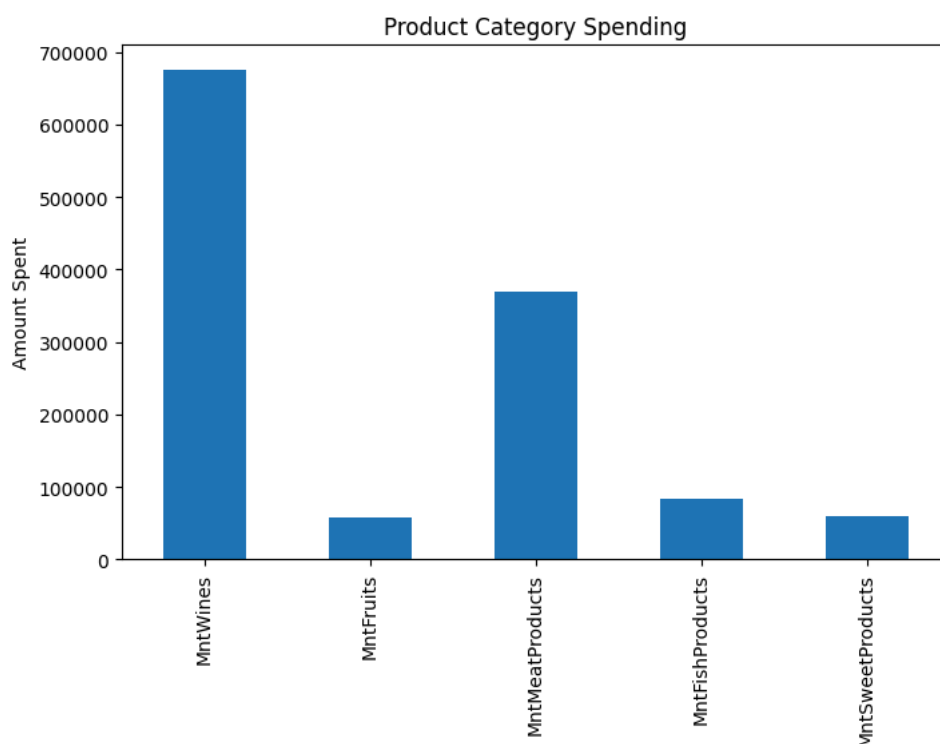
Visualize customer response to marketing campaigns

```
response_counts = df['Response'].value_counts()  
  
plt.figure(figsize=(5,5))  
  
plt.pie(  
    response_counts,  
    labels=response_counts.index,  
    autopct='%1.1f%%'  
)  
  
plt.title('Campaign Response')  
  
plt.show()
```



✓ Analyze spending across product categories

```
products = [  
    'MntWines',  
    'MntFruits',  
    'MntMeatProducts',  
    'MntFishProducts',  
    'MntSweetProducts'  
]  
  
df[products].sum().plot(  
    kind='bar',  
    figsize=(8,5)  
)  
  
plt.title('Product Category Spending')  
plt.ylabel('Amount Spent')  
  
plt.show()
```



Business Insights

1. Customers with higher income tend to spend more overall.
2. Web purchases are higher than catalog purchases, showing strong online engagement.
3. Wine and meat products contribute significantly to customer spending.
4. Campaign response rate is relatively low, suggesting scope for better targeting.
5. High-value customers should be targeted with personalized marketing campaigns.

Recommendations:

- Increase budget for online/web marketing campaigns because web purchases are highest.
- Focus on premium customers with high income levels.
- Improve campaign targeting strategies to increase response rate.
- Promote high-performing product categories like wines and meat products.
- Reduce spending on low-performing marketing channels.

Conclusion:

The EDA revealed that customer income strongly influences spending behavior. Online channels generate high engagement and purchase activity, making them the best target for future marketing investments. Personalized campaigns and optimized digital marketing strategies can improve ROI and business growth.